Heritability

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1 REML-based heritability estimation

These derivations are based on the Methods of Yang et al. [**?**].

1.1 Phenotype model

We can define a quantitative phenotype vector $\mathbf{y}$ as:

$$\mathbf{y} = \mathbf{X}_c\boldsymbol{\beta} + \boldsymbol{\epsilon}$$

Where:

- $\mathbf{y} \in \mathbb{R}^N$: phenotypes. Centered so that $\mathrm{E}[\mathbf{y}] = 0$.
  - N: number of samples.
- $\mathbf{X}_c \in \mathbb{R}^{N\times M_c}$: normalized genotypes for causal variants.
  - M_c: number of causal variants.
  - Normalized according to $\mathbf{X}_{c,i} = \frac{\mathbf{X}'_{c,i} - 2f_i}{\sqrt{2f_i(1-f_i)}}$.
    - $\mathbf{X}'_c \in \{0,1,2\}^{N\times M_c}$: allele dosages.
    - f_i: true population allele frequency for variant i.
    - Such that for each row (variant), $\mathrm{E}[\mathbf{X}_{c,i}] = 0$ and $\mathrm{Var}[\mathbf{X}_{c,i}] = 1$.
- $\boldsymbol{\beta} \in \mathbb{R}^{M_c}$: per-normalized-genotype causal effects.
  - Assume infinitesimal model.
  - Drawn from $\boldsymbol{\beta} \sim \mathcal{N}(\mathbf{0}, \sigma^2_\beta \mathbf{I})$.
    - σ^2_β: variance of causal effects.
    - $\mathbf{I} \in \mathbb{R}^{M_c\times M_c}$: identity matrix.
- $\boldsymbol{\epsilon} \in \mathbb{R}^N$: residual effects (i.e. error or noise term).
  - Drawn from $\boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \sigma^2_\epsilon \mathbf{I})$.
    - σ^2_ϵ: residual variance.
    - $\mathbf{I} \in \mathbb{R}^{N\times N}$: identity matrix.

We assume $\mathbf{X}_c$, $\boldsymbol{\beta}$, and $\boldsymbol{\epsilon}$ are all independent from each other.

1.2 Variance of the phenotype

By making use of the independence between terms, we can define the variance-covariance matrix of $\mathbf{y}$ as:

$$\begin{aligned}
\mathrm{Var}[\mathbf{y}] &= \mathrm{Var}[\mathbf{X}_c\boldsymbol{\beta} + \boldsymbol{\epsilon}] \\
&= \mathrm{Var}[\mathbf{X}_c\boldsymbol{\beta}] + \mathrm{Var}[\boldsymbol{\epsilon}] \\
&= \mathrm{Var}[\mathbf{X}_c]\mathrm{Var}[\boldsymbol{\beta}] + \mathrm{Var}[\boldsymbol{\epsilon}] \\
&= \sigma_\beta^2(\mathbf{X}_c\mathbf{X}_c^\intercal) + \sigma_\epsilon^2\mathbf{I} \\
&= M_c\sigma_\beta^2\frac{\mathbf{X}_c\mathbf{X}_c^\intercal}{M_c} + \sigma_\epsilon^2\mathbf{I} \\
&= \sigma_g^2\mathbf{G} + \sigma_\epsilon^2\mathbf{I}
\end{aligned}$$

where we define $\mathbf{G} = \frac{\mathbf{X}_c\mathbf{X}_c^\intercal}{M_c}$, with $\mathbf{G} \in \mathbb{R}^{N\times N}$, as the genetic relationship matrix (GRM) between individuals. The G_{ii} element is the variance of individual i's normalized genotype vector, while the G_{ij} element is the covariance of individuals i and j's normalized genotype vectors.

We also define $\sigma_g^2 = M_c\sigma_\beta^2$ to be the variance of the total additive genetic effects on the phenotype. We can therefore think of individuals' phenotypes as being derived from the sum of a genetic random effect $\mathbf{g} = \mathbf{X}_c\boldsymbol{\beta}$ and a residual random effect $\boldsymbol{\epsilon}$. From the general rule of multivariable statistics that $\mathrm{Var}[\mathbf{A}\mathbf{v}] = \mathbf{A}\mathrm{Var}[\mathbf{v}]\mathbf{A}^\intercal$, we can write the genetic random effect as coming from a normal distribution:

$$\mathbf{g} \sim \mathcal{N}(\mathrm{E}[\mathbf{X}\boldsymbol{\beta}], \mathrm{Var}[\mathbf{X}\boldsymbol{\beta}])$$

Because $\mathrm{E}[\boldsymbol{\beta}] = \mathbf{0}$, then $\mathrm{E}[\mathbf{X}\boldsymbol{\beta}] = \mathbf{0}$. As for the variance, we can reuse the definitions from earlier, $\mathrm{Var}[\mathbf{X}\boldsymbol{\beta}] = \mathbf{G}\sigma_g^2$. Therefore,

$$\mathbf{g} \sim \mathcal{N}(\mathbf{0}, \mathbf{G}\sigma_g^2)$$

Narrow-sense heritability is defined as the proportion of phenotypic variance, σ_P^2, explained by additive genetic effects:

$$h^2 = \frac{\sigma_g^2}{\sigma_P^2} = \frac{\sigma_g^2}{\sigma_g^2 + \sigma_\epsilon^2}$$

1.3 Estimating the GRM

In practice, we likely do not know the exact set of causal variants and instead must estimate the GRM using a set of genotyped SNPs:

$$\mathbf{A} = \frac{\mathbf{X}\mathbf{X}^\intercal}{M}$$

Where $\mathbf{A}$ is the estimated GRM, $\mathbf{X}$ is the normalized genotype matrix of our genotyped SNPs, and M is the number of genotyped SNPs. Note that because we are also working with a sample, X is normalized using sample allele frequencies, $\mathbf{p}$:

$$\mathbf{X}_i = \frac{\mathbf{X}'_i - 2p_i}{\sqrt{2p_i(1-p_i)}}$$

However, this equation for $\mathbf{A}$ ignores the sampling error associated with each SNP. Let's consider the covariance computation between two individuals for SNP i, which is then summed across M SNPs to get the value for A_{jk}. When $j \neq k$:

$$\begin{aligned}
A_{ijk} &= x_{ij}x_{ik} \\
&= \frac{x'_{ij} - 2p_i}{\sqrt{2p_i(1-p_i)}} \frac{x'_{ik} - 2p_i}{\sqrt{2p_i(1-p_i)}} \\
&= \frac{(x'_{ij} - 2p_i)(x'_{ik} - 2p_i)}{2p_i(1-p_i)}
\end{aligned}$$

Because x'_{ij} and x'_{ik} are independent from each other, the expected value of this is:

$$\begin{aligned}
\mathrm{E}[A_{ijk}] &= \mathrm{E}[\frac{(x'_{ij} - 2p_i)(x'_{ik} - 2p_i)}{2p_i(1-p_i)}] \\
&= \frac{(\mathrm{E}[x'_{ij}] - 2p_i)(\mathrm{E}[x'_{ik}] - 2p_i)}{2p_i(1-p_i)} \\
&= \frac{(2p_i - 2p_i)(2p_i - 2p_i)}{2p_i(1-p_i)} \\
&= 0
\end{aligned}$$

This makes sense if our sample is of unrelated individuals. If the raw genotypes of two individuals at a SNP are independent from each other, then the covariance of their adjusted genotypes should also be zero. Furthermore, the variance of A_{ijk} is given by:

$$\begin{aligned}
\mathrm{Var}[A_{ijk}] &= \mathrm{Var}[\frac{(x'_{ij} - 2p_i)(x'_{ik} - 2p_i)}{2p_i(1-p_i)}] \\
&= \frac{\mathrm{Var}[(x'_{ij} - 2p_i)]\mathrm{Var}[(x'_{ik} - 2p_i)]}{(2p_i(1-p_i))^2} \\
&= \frac{\mathrm{Var}[(x'_{ij})]\mathrm{Var}[(x'_{ik})]}{(2p_i(1-p_i))^2} \\
&= \frac{(2p_i(2-p_i))(2p_i(2-p_i))}{(2p_i(1-p_i))^2} \\
&= 1
\end{aligned}$$

So, the variance in A_{jk} is independent of allele frequency, which is a desirable property.

But do these properties hold up when $j = k$ (i.e. variance of an indvidual's genotype)?

$$\begin{aligned}
A_{ijj} &= x_{ij}^2 \\
&= \Big(\frac{x'_{ij} - 2p_i}{\sqrt{2p_i(1-p_i)}}\Big)^2 \\
&= \frac{(x'_{ij} - 2p_i)^2}{2p_i(1-p_i)}
\end{aligned}$$

For deriving $\mathrm{E}[A_{ijj}]$, we make use of:

$$\begin{aligned}
E[{x'}_{ij}^2] &= \sum_{n=0}^{2} \mathrm{E}[{x'}_{ij}^2 | {x'}_{ij} = n] \Pr({x'}_{ij} = n) \\
&= (0)^2(1-p_i)^2 + (1)^2 2p_i(1-p_i) + (2)^2(p_i)^2 \\
&= 2p_i - 2p_i^2 + 4p_i^2 \\
&= 2p_i^2 + 2p_i
\end{aligned}$$

Allowing us to calculate $\mathrm{E}[A_{ijj}]$:

$$\begin{aligned}
\mathrm{E}[A_{ijj}] &= \mathrm{E}\Big[\frac{(x'_{ij} - 2p_i)^2}{2p_i(1-p_i)}\Big] \\
&= \frac{\mathrm{E}[{x'}_{ij}^2 - 4p_i x'_{ij} + 4p_i^2]}{2p_i(1-p_i)} \\
&= \frac{\mathrm{E}[{x'}_{ij}^2] - 4p_i \mathrm{E}[x'_{ij}] + 4p_i^2}{2p_i(1-p_i)} \\
&= \frac{2p_i^2 + 2p_i - 4p_i(2p_i) + 4p_i^2}{2p_i(1-p_i)} \\
&= \frac{2p_i - 2p_i^2}{2p_i(1-p_i)} \\
&= \frac{2p_i(1-p_i)}{2p_i(1-p_i)} \\
&= 1
\end{aligned}$$

So, the expected variance of an individual's normalized genotype is 1 and independent of the allele frequency, which is a desirable property. However, this frequency-independence does not hold for the variance. For simplicity, let's denote $H = 2p_i(1-p_i)$ and make use of $\mathrm{Var}[Y] = \mathrm{E}[Y^2] - \mathrm{E}[Y]^2$:

$$\begin{aligned}
\mathrm{Var}[A_{ijj}] &= \mathrm{Var}[\frac{(x'_{ij} - 2p_i)^2}{H}] \\
&= \frac{\mathrm{Var}[(x'_{ij} - 2p_i)^2]}{(H)^2} \\
&= \frac{\mathrm{E}[((x'_{ij} - 2p_i)^2)^2] - \mathrm{E}[(x'_{ij} - 2p_i)^2)]^2}{(H)^2} \\
&= \frac{(H) - (H)^2}{(H)^2} \\
&= \frac{(H)(1 - H)}{(H)^2} \\
&= \frac{1 - H}{H} \\
&= \frac{1 - 2p_i(1 - p_i)}{2p_i(1 - p_i)}
\end{aligned}$$

The full derivation for why $\mathrm{E}[((x'_{ij} - 2p_i)^2)^2] = \mathrm{E}[(x'_{ij} - 2p_i)^2] = 2p_i(1 - p_i)$ is very lengthy algebraically, but can be shortcutted by using the formula for the higher moments of a binomially distributed variable, where $n = 2$ and $p = p_i$ [**?**]. Importantly, the variance of A_{jj} therefore depends on the allele frequencies of the SNPs, even after normalization. **A** would be a better estimator of the GRM if for the same individual, its variance did not depend on the allele frequency, so we want to adjust A_{ijj} so that $\mathrm{E}[A_{ijj}] = 1$ like before, but also that $\mathrm{Var}[A_{ijj}] = 1$ like for unrelated individuals.

To be frank, I am not sure how the authors derived it, but this equation holds both properties:

$$A_{ijj} = 1 + \frac{x'^2_{ij} - (1 + 2p_i)x_{ij} + 2p_i^2}{2p_i(1 - p_i)}$$

We now explicitly define **A** as follows:

$$A_{jk} = \frac{1}{M}\sum_{i=1}^{M} A_{ijk} = \begin{cases} \frac{1}{M}\sum_{i=1}^{M} \frac{(x'_{ij}-2p_i)(x'_{ik}-2p_i)}{2p_i(1-p_i)}, & j \neq k \\ 1 + \frac{1}{M}\sum_{i=1}^{M} \frac{x'^2_{ij}-(1+2p_i)x'_{ij}+2p_i^2}{2p_i(1-p_i)}, & j = k \end{cases}$$

1.4 Estimating the genetic variance component through REML

Notice that earlier, we were able to define the phenotype $\mathbf{y}$ as a sum of random effects, treating **A** as a good approximation of **G**:

$$\mathbf{y} = \mathbf{g} + \boldsymbol{\epsilon} \quad \text{where} \quad \mathbf{g} \sim \mathcal{N}(\mathbf{0}, \sigma_g^2\mathbf{A}) \quad \text{and} \quad \boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \sigma_\epsilon^2\mathbf{I})$$

The unknown parameters here, σ_g^2 and σ_ϵ^2, can be estimated through REML (Restricted Maximum Likelihood; since this model doesn't include any fixed effects, simple Maximum Likelihood would also suffice). See the entry on "Maximum Likelihood" for a derivation of how variance component parameters are estimated through REML.

1.5 Heritability of a polygenic risk score

Here, we consider a polygenic score (PGS) $\hat{y}_i$ that acts to predict the trait value y_i of an individual. We assume the PGS only captures signal through the genetic component g_i of the trait and not through the random noise ϵ_i. That means we can model a polygenic score (PGS) as follows:

$$\hat{\mathbf{y}} = \mathbf{g} + \boldsymbol{\zeta}$$

where $\boldsymbol{\zeta} \in \mathbb{R}^N$ is random noise in the PGS: $\boldsymbol{\zeta} \sim \mathcal{N}(\mathbf{0}, \sigma_\zeta^2 \mathbf{I})$. PGS accuracy is commonly defined by its R^2 value:

$$\mathrm{R}^2 = \mathrm{Corr}[\mathbf{y}, \hat{\mathbf{y}}]^2$$

Under our assumptions, $\mathrm{R}^2 \leq h^2$. If the phenotype has been standardized so that $\sigma_P^2 = 1$, then $h^2 = \sigma_g^2$ and $\mathrm{R}^2 \leq \sigma_g^2$. We can derive the value of σ_ζ^2 as follows:

$$\begin{aligned}
\mathrm{R}^2 &= \mathrm{Corr}[\mathbf{y}, \hat{\mathbf{y}}]^2 \\
&= \frac{\mathrm{Cov}[\mathbf{y}, \hat{\mathbf{y}}]^2}{\mathrm{Var}[\mathbf{y}]\mathrm{Var}[\hat{\mathbf{y}}]} \\
&= \frac{\mathrm{Cov}[\mathbf{g} + \boldsymbol{\epsilon}, \mathbf{g} + \boldsymbol{\zeta}]^2}{\mathrm{Var}[\mathbf{g} + \boldsymbol{\epsilon}]\mathrm{Var}[\mathbf{g} + \boldsymbol{\zeta}]}
\end{aligned}$$

Because $\mathbf{g}$, $\boldsymbol{\epsilon}$, and $\boldsymbol{\zeta}$ are independent from each other:

$$\begin{aligned}
\mathrm{Cov}[\mathbf{g} + \boldsymbol{\epsilon}, \mathbf{g} + \boldsymbol{\zeta}] &= \mathrm{Cov}[\mathbf{g}, \mathbf{g}] + \mathrm{Cov}[\mathbf{g}, \boldsymbol{\zeta}] + \mathrm{Cov}[\boldsymbol{\epsilon}, \mathbf{g}] + \mathrm{Cov}[\boldsymbol{\epsilon}, \boldsymbol{\zeta}] \\
&= \mathrm{Var}[\mathbf{g}] \\
&= \sigma_g^2
\end{aligned}$$

and

$$\begin{aligned}
\mathrm{Var}[\mathbf{y}] &= \mathrm{Var}[\mathbf{g} + \boldsymbol{\epsilon}] = \sigma_g^2 + \sigma_\epsilon^2 = \sigma_P^2 \\
\mathrm{Var}[\hat{\mathbf{y}}] &= \mathrm{Var}[\mathbf{g} + \boldsymbol{\zeta}] = \sigma_g^2 + \sigma_\zeta^2
\end{aligned}$$

Therefore,

$$\begin{aligned}
\mathrm{R}^2 &= \frac{(\sigma_g^2)^2}{(\sigma_g^2 + \sigma_\epsilon^2)(\sigma_g^2 + \sigma_\zeta^2)} \\
&= \frac{(\sigma_g^2)^2}{\sigma_P^2(\sigma_g^2 + \sigma_\zeta^2)}
\end{aligned}$$

Solving this for σ_ζ^2:

$$\begin{aligned}
\mathrm{R}^2 \sigma_P^2 (\sigma_g^2 + \sigma_\zeta^2) &= (\sigma_g^2)^2 \\
\sigma_g^2 + \sigma_\zeta^2 &= \frac{(\sigma_g^2)^2}{\mathrm{R}^2 \sigma_P^2} \\
\sigma_\zeta^2 &= \frac{(\sigma_g^2)^2}{\mathrm{R}^2 \sigma_P^2} - \sigma_g^2 \\
\sigma_\zeta^2 &= \sigma_g^2 \left(\frac{\sigma_g^2}{\mathrm{R}^2 \sigma_P^2} - 1 \right)
\end{aligned}$$

If we are working with a standardized phenotype, then $\sigma_P^2 = 1$ and this simplifies to:

$$\sigma_\zeta^2 = \sigma_g^2 \left(\frac{\sigma_g^2}{\mathrm{R}^2} - 1 \right)$$

This also makes clear why R^2 cannot be larger than the heritability under this model. If $\sigma_\zeta^2 = 0$, the PGS is exactly the genetic component, so $\hat{\mathbf{y}} = \mathbf{g}$ and $\mathrm{R}^2 = h^2$. Adding PGS noise increases $\mathrm{Var}[\hat{\mathbf{y}}]$ without increasing $\mathrm{Cov}[\mathbf{y}, \hat{\mathbf{y}}]$, so R^2 decreases. This also means that:

$$\mathrm{Var}[\hat{\mathbf{y}}] = \frac{(\sigma_g^2)^2}{\mathrm{R}^2 \sigma_P^2} = \frac{\sigma_g^2 h^2}{\mathrm{R}^2}$$

Suppose that the PGS is then scaled $\hat{\mathbf{y}}' = c\hat{\mathbf{y}}$ such that $\mathrm{Var}[\hat{\mathbf{y}}'] = 1$, where:

$$c^2 = \frac{1}{\mathrm{Var}[\hat{\mathbf{y}}]} = \frac{\mathrm{R}^2 \sigma_P^2}{(\sigma_g^2)^2}$$

We can then write the scaled PGS as:

$$\begin{aligned} \hat{\mathbf{y}}' &= c(\mathbf{g} + \boldsymbol{\zeta}) \\ &= c\mathbf{g} + c\boldsymbol{\zeta} \end{aligned}$$

The variance explained by the genetic component in the scaled PGS is:

$$\begin{aligned} \mathrm{Var}[c\mathbf{g}] &= c^2 \mathrm{Var}[\mathbf{g}] \\ &= \frac{\mathrm{R}^2 \sigma_P^2}{(\sigma_g^2)^2} \sigma_g^2 \\ &= \frac{\mathrm{R}^2 \sigma_P^2}{\sigma_g^2} \\ &= \frac{\mathrm{R}^2}{h^2} \end{aligned}$$

and the variance explained by the residual component in the scaled PGS is:

$$\begin{aligned} \mathrm{Var}[c\boldsymbol{\zeta}] &= c^2 \mathrm{Var}[\boldsymbol{\zeta}] \\ &= \frac{\mathrm{R}^2 \sigma_P^2}{(\sigma_g^2)^2} \left[\frac{(\sigma_g^2)^2}{\mathrm{R}^2 \sigma_P^2} - \sigma_g^2 \right] \\ &= 1 - \frac{\mathrm{R}^2 \sigma_P^2}{\sigma_g^2} \\ &= 1 - \frac{\mathrm{R}^2}{h^2} \end{aligned}$$

Therefore, if we fit the following REML model and treat $\mathbf{A}$ as a good approximation of the true GRM:

$$\hat{\mathbf{y}}' \sim \mathcal{N}(\mathbf{0}, \sigma_1^2 \mathbf{A} + \sigma_2^2 \mathbf{I})$$

the expected values of σ_1^2 and σ_2^2 are:

$$\begin{aligned} \mathrm{E}[\sigma_1^2] &= \frac{\mathrm{R}^2 \sigma_P^2}{\sigma_g^2} = \frac{\mathrm{R}^2}{h^2} \\ \mathrm{E}[\sigma_2^2] &= 1 - \frac{\mathrm{R}^2 \sigma_P^2}{\sigma_g^2} = 1 - \frac{\mathrm{R}^2}{h^2} \end{aligned}$$

In other words, the heritability of the scaled PGS is the ratio between PGS accuracy and trait heritability.